

Track 2: mechanism before treatment

0:00-0:18

Start from a scorer-confirmed genetic ranking, not from a drug list

100 / 100

rank points

F-max **1.000**

EPCR **0.78**

BUB1B L737* + N1002K

What this validates

VALIDATED

Genetic ranking used as the Track 2 input

What this does NOT validate

UNRESOLVED

Phase cis or trans?

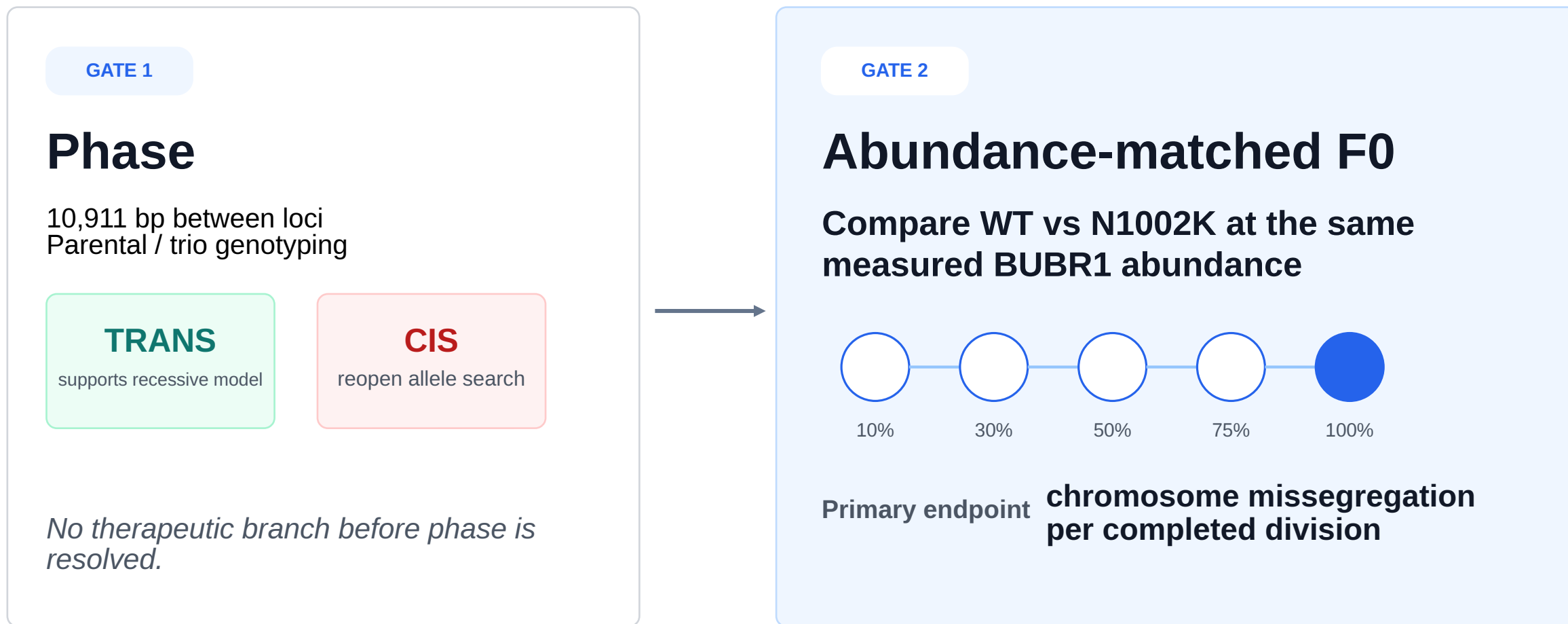
Mechanism what does N1002K do?

Therapy does anything rescue it?

Two gates before any drug test

0:18-0:40

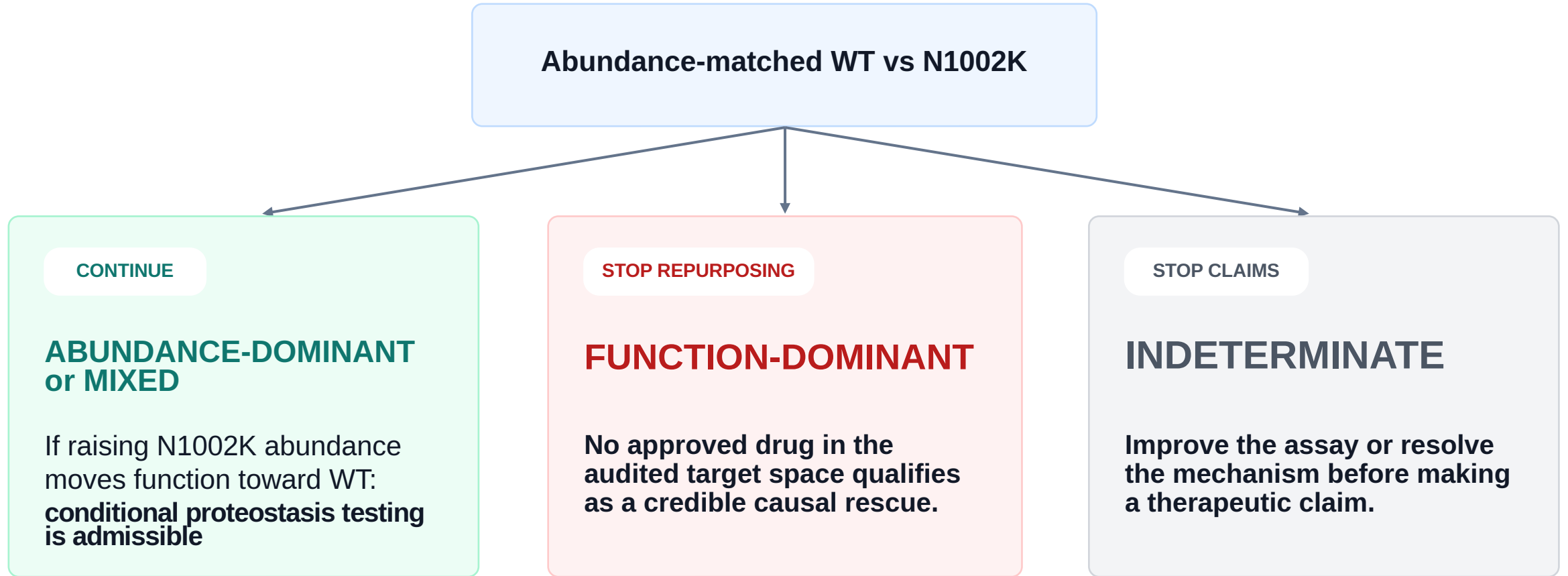
Phase first. Then measure N1002K function at matched BUBR1 abundance.



The mechanism decides whether repurposing continues

0:40-1:02

A negative mechanism result is allowed to stop the drug story.



This is a real gate, not a decorative branch.

The drug-coverage audit was repaired

1:02-1:27

Exact UniProt-component mapping, approval filtering, and explicit NOT_RUN states

13

prespecified targets

14

unique approved
ChEMBL parent drugs

11

approved DrugCentral
structures

NO NUMERICAL WINNER

coverage is audited; direction is adjudicated

Desired rescue axes remain sparse

HSF1 **0 / 0** ChEMBL / DrugCentral

SIRT2 **0 / 0** ChEMBL / DrugCentral

PPP2CA **0 / 0** ChEMBL / DrugCentral

Direct / downstream targets are well represented

PSMB5 **3 / 2**

JAK1 **8 / 4**

JAK2 **9 / 4**

MTOR **1 / 2**

Lesson: annotation density is not therapeutic credibility.

Approval is not enough

1:27-1:57

Regulatory status plus exposure plausibility decide whether testing is credible.

US FDA-approved in NPC

EU CHMP negative opinion, July 2026; re-exam ongoing

Recommended-regimen pediatric Cmax

523 ng/mL

≈ 1.67 μM

free-base equivalent; ~1.50 μM unbound

Some motivating nonclinical experiments

100-400 μM



DOES NOT PROVE INEFFICACY

It changes the experiment: start in the low-μM clinically anchored range.

Pre-commitment before rescue claims

NB17 fixes evidence strength; NB10 tests false rescue.

NB17 evidence ceiling

6 B/LB + 5 P/LP

Perfect separation Moderate only

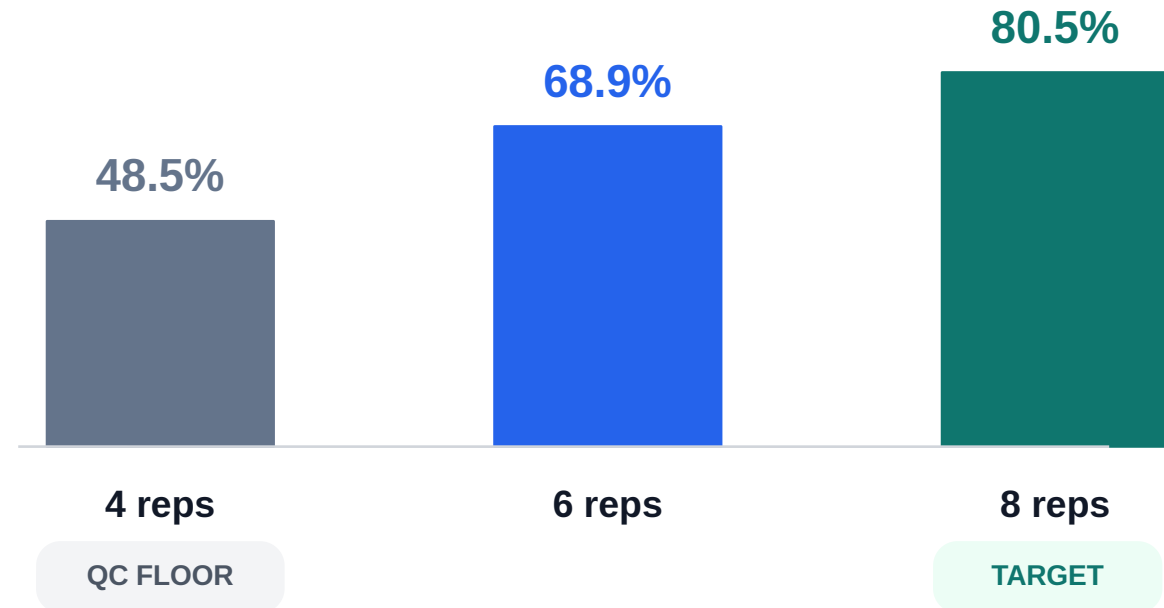
Strong evidence prospectively
inadmissible

Threshold locked from controls
before N1002K unblinding.

Any single control loss leaves
<11 controls.

NB10 synthetic false-rescue audit

Decision threshold: $\geq 25\%$ RRR. Scenario: 40% true RRR. Cytostasis/kill: 0 clean rescues.



4 reps = QC floor. Target 8. Report effect size + 95% CI.

For the family and clinical team

What is known, what is not, and the cheapest decisive next test

HF: kotgienia

KNOWN

What is known

BUB1B L737* + N1002K is the scorer-confirmed leading genetic hypothesis.

UNKNOWN

What is not known

Phase, N1002K mechanism, and therapy are not proven.

NEXT

Cheapest decisive test

Parental genotyping of both BUB1B variants: trans strengthens the model; cis reopens the search.

NO TREATMENT RECOMMENDATION

Track 2 delivers a pre-specified way to decide when to continue and when to stop.